Quarto Story Telling

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Storytelling in R

Introduction

Data interpretation and storytelling mean understanding the results of data analysis, answering important questions, and sharing the findings with others. Visual tools like graphs, charts, plots, and dashboards are often used to make the information clearer and easier to explain. In this chapter, we will look at different data analysis tools and techniques, as well as the different types of data analytics.

A version of this textbook is available at https://nxrolinf.github.io/QuartoSTell

Lab 3: Data Storytelling and Visualization in R

Overview

This lab explores data storytelling using R through: - Exploratory data analysis (EDA) - Transformations and skewness - Regression visualization - Time series anomalies - Composite plots - Interactive visualization (Shiny) - A capstone narrative

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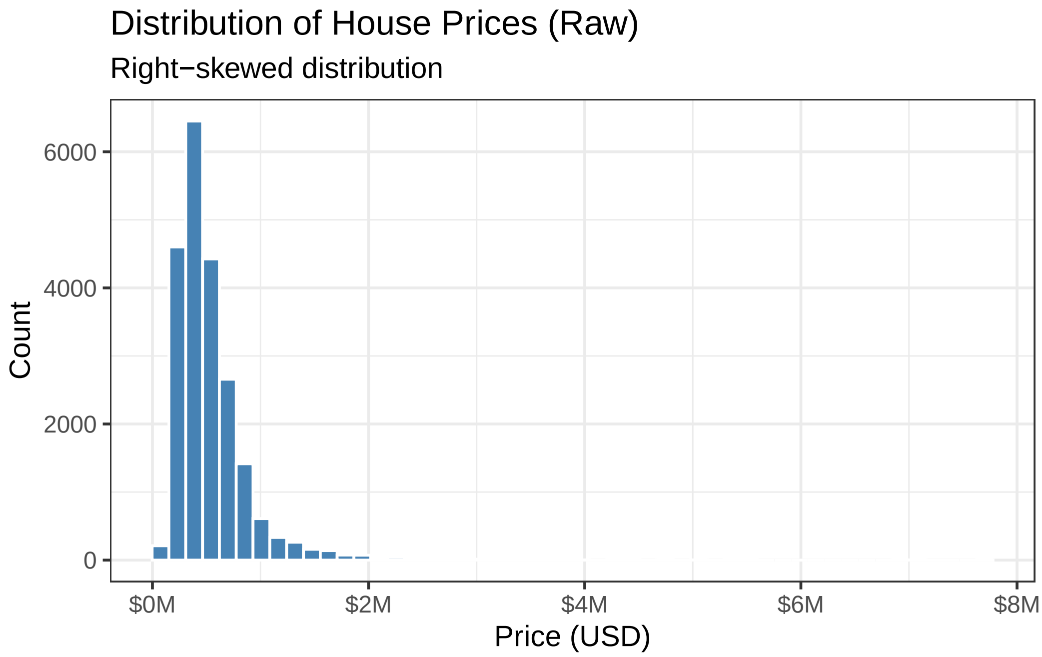
Setup

```r
# Install if needed
# install.packages(c("tidyverse", "moderndive", "fivethirtyeight",
#                    "patchwork", "scales", "shiny"))

library(tidyverse)
library(moderndive)
library(fivethirtyeight)
library(patchwork)
library(scales)
```

Exercise 1: Exploratory Data Analysis & Skewness

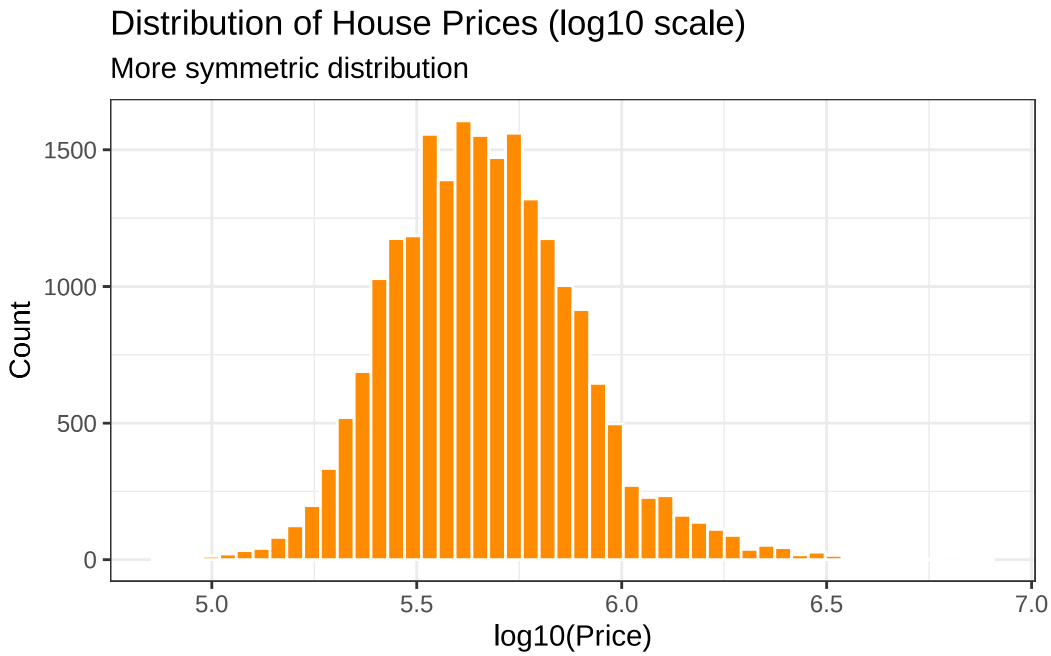
Raw Distribution

```
data("house_prices")

p1_raw <- ggplot(house_prices, aes(x = price)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  scale_x_continuous(labels = label_dollar(scale = 1e-6, suffix = "M")) +
  labs(
    title = "Distribution of House Prices (Raw)",
    subtitle = "Right-skewed distribution",
    x = "Price (USD)",
    y = "Count"
  ) +
  theme_bw()

p1_raw
```



Log Transformation

```
house_prices <- house_prices %>%
  mutate(
    log10_price = log10(price),
    log10_size  = log10(sqft_living)
  )

p1_log <- ggplot(house_prices, aes(x = log10_price)) +
  geom_histogram(bins = 50, fill = "darkorange", color = "white") +
  labs(
    title = "Distribution of House Prices (log10 scale)",
    subtitle = "More symmetric distribution",
    x = "log10(Price)",
    y = "Count"
  ) +
  theme_bw()

p1_log
```



Interpretation

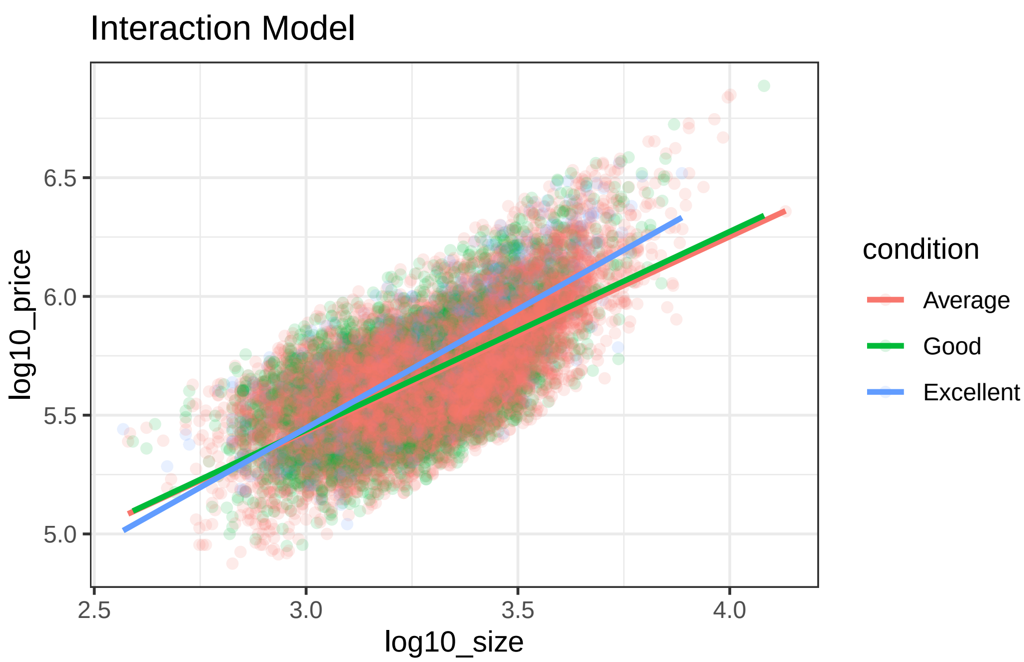
The raw distribution is heavily right-skewed due to high-value outliers.
The log transformation reveals a more symmetric, near-normal distribution, improving interpretability.

Exercise 2: Interaction vs Parallel Slopes

```
house_prices <- house_prices %>%
  mutate(condition = factor(condition, levels = 1:5,
                            labels = c("Poor","Fair","Average","Good","Excellent")))

house_sub <- house_prices %>%
  filter(condition %in% c("Average", "Good", "Excellent"))
```

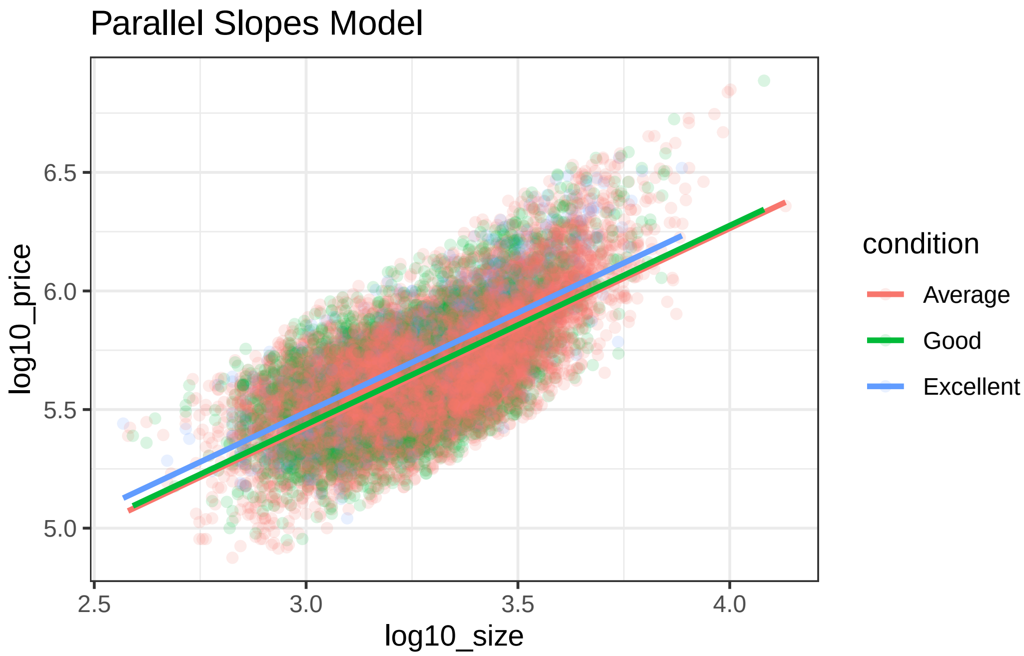
Interaction Model

```
p2_interaction <- ggplot(house_sub,
                         aes(x = log10_size, y = log10_price, color = condition)) +
  geom_point(alpha = 0.15) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Interaction Model") +
  theme_bw()

p2_interaction
```



Parallel Slopes Model

```
p2_parallel <- ggplot(house_sub,
                      aes(x = log10_size, y = log10_price, color = condition)) +
  geom_point(alpha = 0.15) +
  moderndive::geom_parallel_slopes(se = FALSE) +
  labs(title = "Parallel Slopes Model") +
  theme_bw()

p2_parallel
```



Interpretation

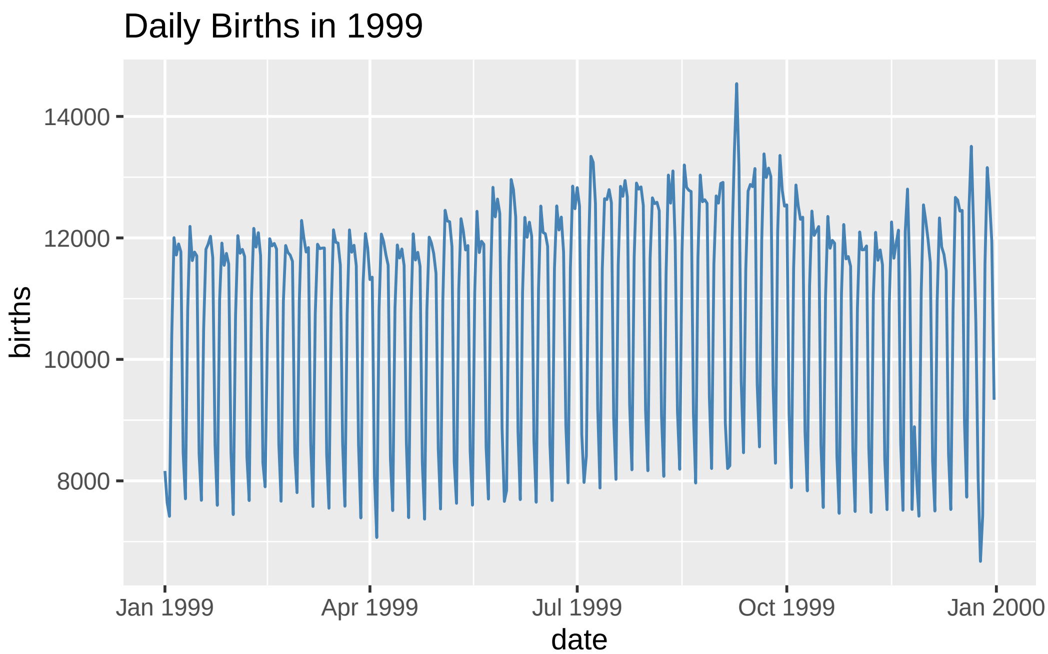
Slopes are similar across conditions, supporting a simpler parallel slopes interpretation: - Size affects price consistently - Condition shifts baseline price

---

Exercise 3: Time Series Anomaly

```
data("US_births_1994_2003")

births_1999 <- US_births_1994_2003 %>%
  filter(year == 1999) %>%
  mutate(date = as.Date(paste(year, month, date_of_month, sep = "-")))
```

```
ggplot(births_1999, aes(x = date, y = births)) +
  geom_line(color = "steelblue") +
  labs(title = "Daily Births in 1999")
```



Peak Detection

```
births_1999 %>%
  arrange(desc(births)) %>%
  head()
```

```
# A tibble: 6 x 6
   year month date_of_month date       day_of_week births
  <int> <int>         <int> <date>     <ord>        <int>
1  1999     9             9 1999-09-09 Thurs        14540
2  1999    12            21 1999-12-21 Tues         13508
3  1999     9             8 1999-09-08 Wed          13437
4  1999     9            21 1999-09-21 Tues         13384
5  1999     9            28 1999-09-28 Tues         13358
6  1999     7             7 1999-07-07 Wed          13343
```

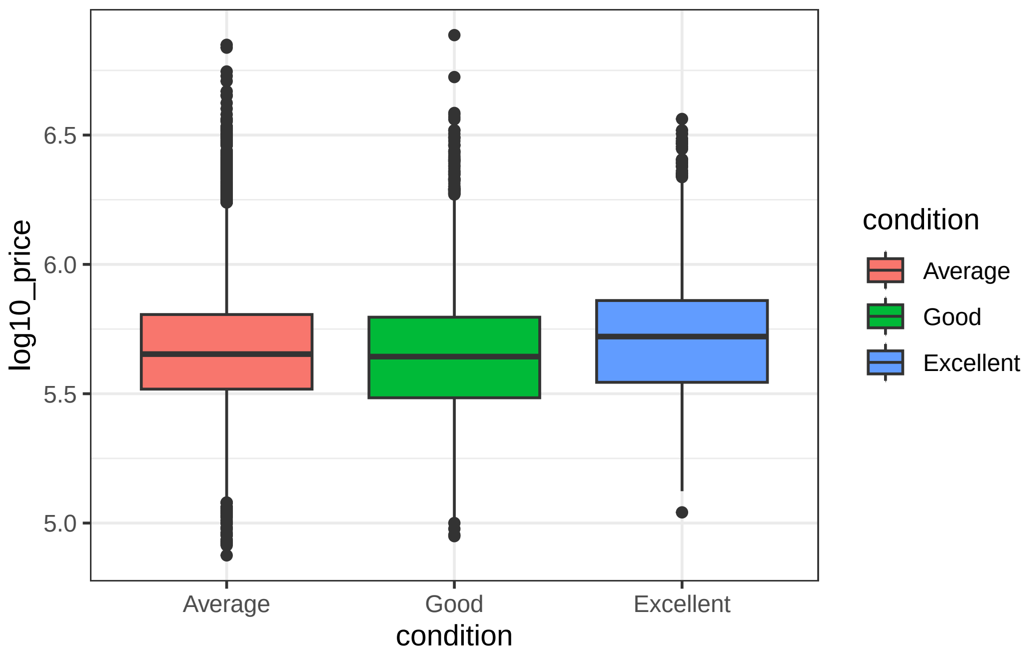
Interpretation

The largest spike occurs on **September 9, 1999 (9/9/99)**, likely due to cultural preferences influencing scheduled births.

Exercise 4: Composite Visualization

Plot A

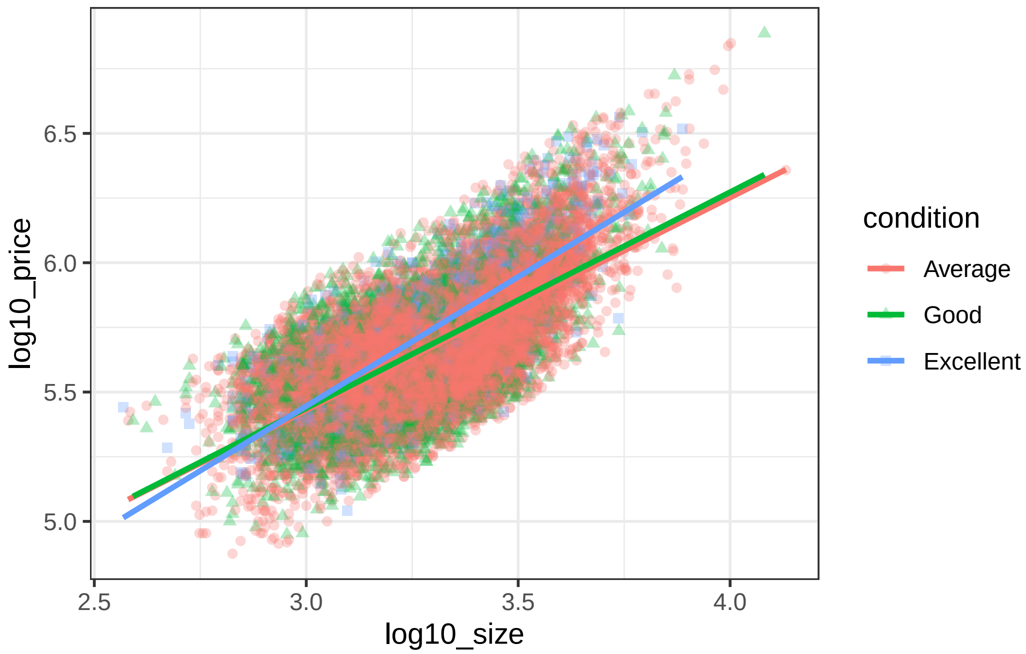
```r
p4a <- ggplot(house_sub,
              aes(x = condition, y = log10_price, fill = condition)) +
  geom_boxplot() +
  theme_bw()
p4a
```



Plot B

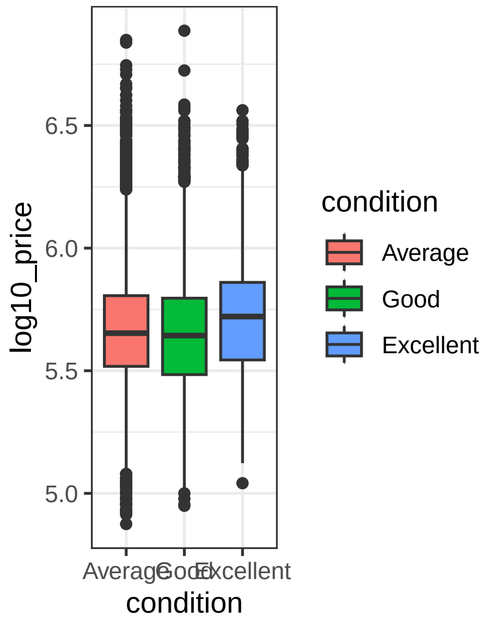
```r
p4b <- ggplot(house_sub,
              aes(x = log10_size, y = log10_price,
                  color = condition, shape = condition)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE) +
```

```
  theme_bw()
p4b
```

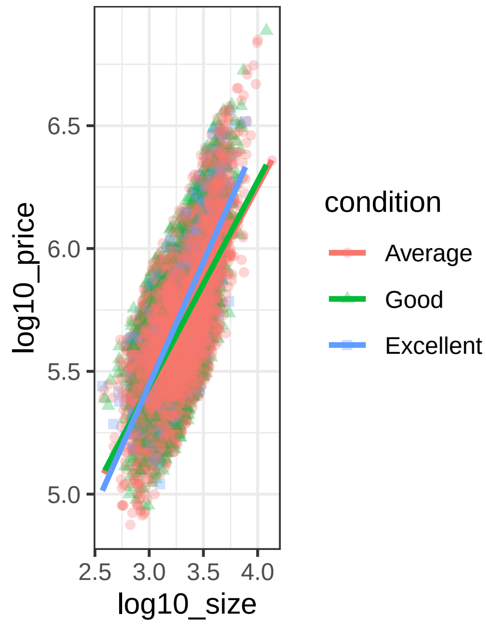


Combined

```
p4a | p4b
```



log10_price
6.5
6.0
5.5
5.0
Average
Good
Excellent
condition
condition
Average
Good
Excellent



log10_price
6.5
6.0
5.5
5.0
2.5
3.0
3.5
4.0
log10_size
condition
Average
Good
Excellent

Exercise 5: Shiny App

Run separately as `app.R`

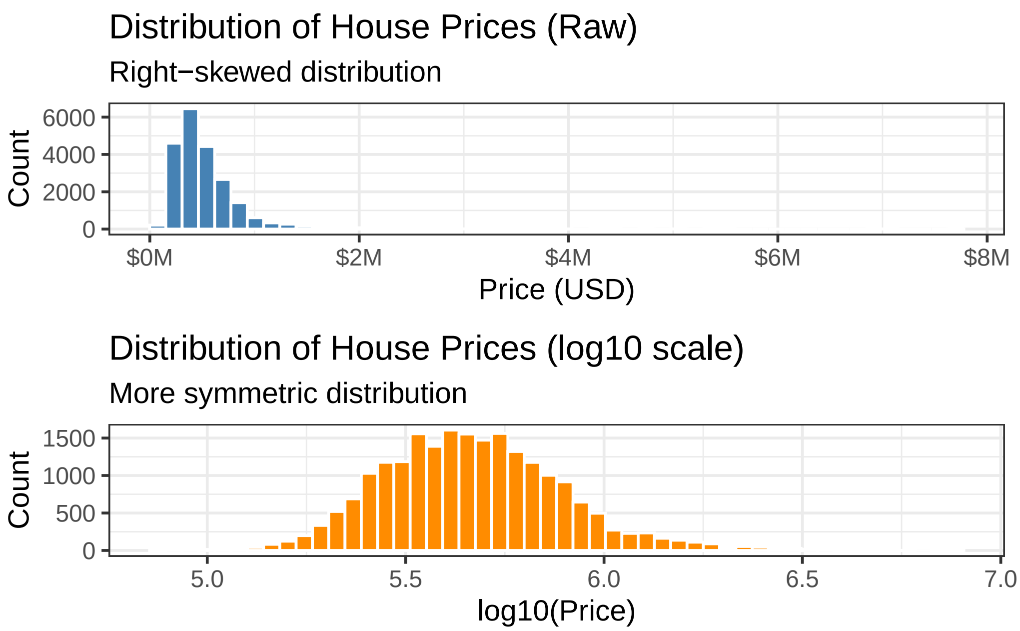
```
# shinyApp(ui, server)
```

(App code omitted in rendering but included in source file)

Capstone: What Drives House Prices?

Step 1: Distribution

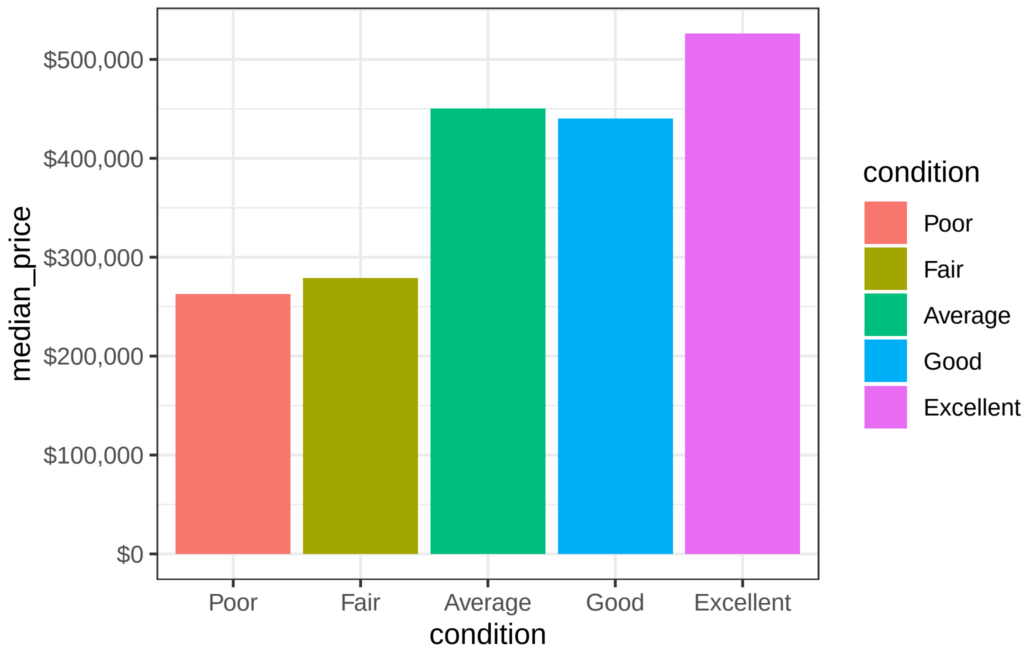
```
cap1 <- p1_raw / p1_log
cap1
```



Step 2: Condition Premium

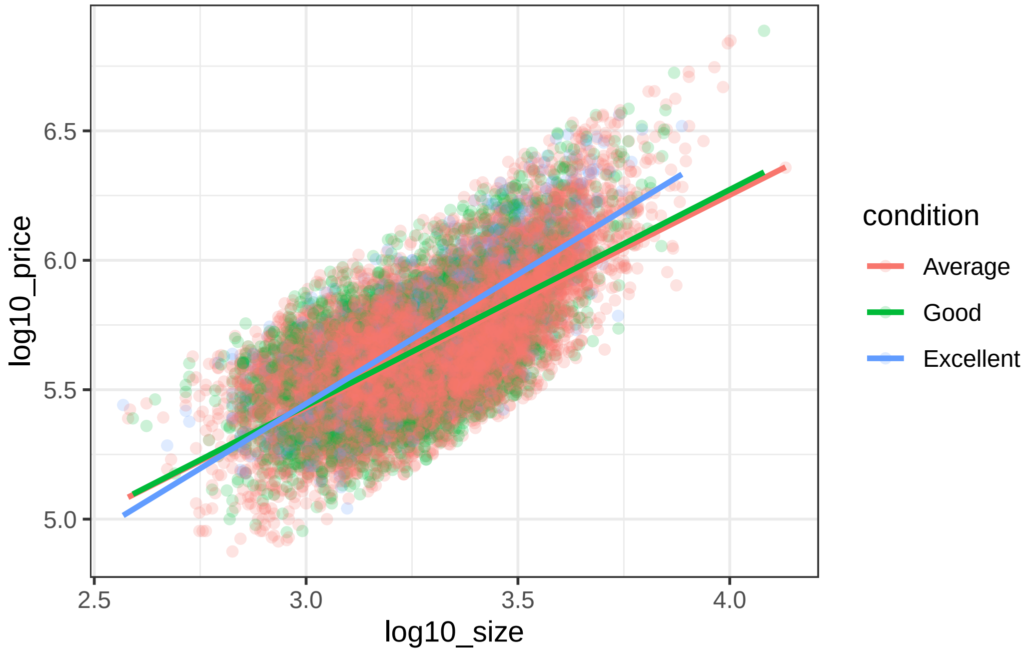
```
median_by_cond <- house_prices %>%
  group_by(condition) %>%
  summarise(median_price = median(price), .groups = "drop")

cap2 <- ggplot(median_by_cond,
               aes(x = condition, y = median_price, fill = condition)) +
  geom_col() +
```

```
  scale_y_continuous(labels = label_dollar()) +
  theme_bw()

cap2
```



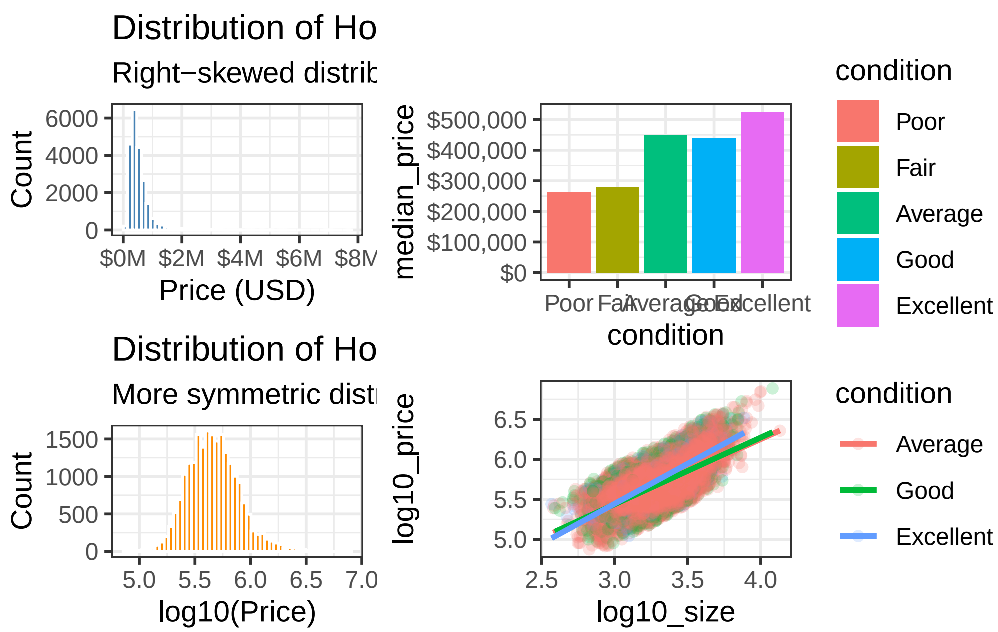
Step 3: Size + Condition

```
cap3 <- ggplot(house_sub,
               aes(x = log10_size, y = log10_price, color = condition)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm", se = FALSE) +
  theme_bw()

cap3
```



Final Narrative

```
cap1 | (cap2 / cap3)
```



Conclusions

- House prices are **right-skewed**, requiring transformation.
- **Condition strongly affects price levels**.
- **Size increases price consistently across conditions**.

Call to Action

- Sellers should invest in condition improvements.
- Buyers can find value in larger homes needing upgrades.

Appendix: Shiny App Code

```r
#! eval: false
library(shiny)

house_shiny <- house_prices %>%
  filter(condition %in% c("Average", "Good", "Excellent"))

ui <- fluidPage(
  titlePanel("House Prices Explorer"),
  sidebarLayout(
    sidebarPanel(
      selectInput("condition_select", "Condition:",
                  choices = c("Average", "Good", "Excellent"))
    ),
    mainPanel(plotOutput("scatter_plot"))
  )
)

server <- function(input, output) {
  filtered_data <- reactive({
    house_shiny %>% filter(condition == input$condition_select)
  })

  output$scatter_plot <- renderPlot({
    ggplot(filtered_data(),
           aes(x = log10_size, y = log10_price)) +
      geom_point(alpha = 0.3) +
      geom_smooth(method = "lm") +
      theme_bw()
  })
}

shinyApp(ui, server)
```

! Work task

For the final part of the lab, you must produce a coherent data story that includes:

1. **At least three different plot types**.
2. **Proper annotation** (titles, axis labels with units, and captions).
3. **A call to action** or a concluding summary of what the visuals reveal.

! Delivery of your task

You're supposed to prepare a qmd document (Quarto, have to install the Quarto CLI from quarto.org). This is a lightweight notebook, fully in R.
Use the **visual** interface for easier development (check the small toolbar, on top of your document window):

1. New file -> Quarto document (you can use any format, as you'll deliver the qmd file), just check that the R kernel is enabled.
2. Organize your document into sections/lists, like any simple HTML dcument.
3. Add your R code, and execute it (insert Executable cell, R), to make sure that you have all the required libraries. If needed, include the necessary libraries at the top of your file, as an executable cell.
   Notice that your libraries come from R (RStudio), don't install them via Quarto documents.
4. Render, and you'll check the output in your browser. Automatic rendering is also available (check the Render on save option).
5. Notice that you can also include simple code chunks, or inhibit code execution (check various insert options, and options for the various cells).
6. If rendering works, you'll see your document (if HTML) in your browser.